UNIT-2

AP & GP

* **Arithmematic Progression**: Difference is same.

$t_n = a + (n-1)d$

f.t. — Total no. of terms — common diff. $= a_1 - a_2$

$S_n = n/2\,[2a + (n-1)d]$

$= \frac{n}{2}\left[a + a + (n-1)d\right] = \frac{n}{2}\left[a + t_n\right]$

Q. 5, 9, 13 . . . , 89

$t_n = \underbrace{\text{Last Term} - \text{first Term}}_{\text{C.D.}} + 1 \Rightarrow \frac{89-5}{4} + 1 = \underline{22}$

Q. −11, −8, −5, . . . t_{10}

$t_{10} = a + (n-1)d = -11 + (10-1) \times 3 = \underline{16}$

$S_{10} = n/2\,[2a + (n-1)d]$

$5\,[-22 + 27] = \underline{25}$

Q. find the sum of first 100 terms of AP.

$\underbrace{1, 4}_{5}, \underbrace{6, 5}_{11}, \underbrace{11, 6}_{16} \ldots$ $a = 5,\ d = 6$

$S_{50} = 25\,[10 + (49) \times 6] = 7600$

Q. find the sum of 99 terms.

$\underbrace{1 - 2 - 3}_{-4}, \underbrace{+2 - 3 - 4}_{-5} + \underbrace{3 - 4 - 5}_{-6} + \ldots t_{33}$

$t_{33} = -4(32) - 1 = \underline{-660}$

$S_n = \frac{33}{2}\left[-4 + (32) - 1\right]$

Q. Find the sum of all two digit no. which will be exactly divided by 9.

Soln. 18, 27, ... 99

Total $= \frac{99-18}{9} + 1 = 10$.

$S_{10} = \frac{10}{2}[18+99] = \underline{585}$

Q. Find the sum of all two digit no. which when div. by 9 leaves a remainder 3.

Soln. 12, 21, ... 93.

$t_n = \frac{93-12}{9} + 1 = 10$

$S_{10} = \frac{10}{2}[a + t_n] = 5[12+93] = \underline{525}$

Q. The sum of 2nd and 5th term of an AP is 8 and that of the 3rd and 7th term is 14. Find the 11th term.

Soln. $T_2 + T_5 = 8$ $\qquad$ $T_3 + T_7 = 14$

$a+d+a+4d = 8$ $\qquad$ $a+2d+a+6d = 14$

$2a + 5d = 8$ —— ① $\qquad$ $2a + 8d = 14$ —— ②

$2a + 8d = 14$

$-\ 2a + 5d = 8$

$3d = 6$ $\qquad$ $d = 2$, $a = -1$.

$a + 10d = -1 + 20 = \boxed{19}$

Q. If $t_1 + t_5 + t_{10} + t_{15} + t_{20} + t_{24} = 225$, find $t_{24} = ?$

Soln. $S_{24} = 12[2a + 23d]$

$a + a + 4d + a + 9d + a + 14d + a + 19d + a + 23d = 225$

$S_{24} = 12[2a + 23d]$ $\qquad$ $6a + 69d = 225$

$12 \times 75 = \underline{900}$ $\qquad$ $2a + 23d = 75$

Q. If $S_n = n^2 + 2n$ then find the first term and c.d.

Soln. $S_n = n/2\,[a + (n-1)d] = n^2 + 2n$.

$S_1 = t_1 = 3$

$S_2 = t_1 + t_2 = 8$ $\quad$ $t_2 = 5$ $\qquad$ $\underline{a = 3,\ d = 2}$

Q. If sum of first 11 terms of an AP = sum of first 19 terms of AP then find the sum of first 30 terms.

Soln. $S_{11} = S_{19}$ $\quad$ $S_{30} = ?$

$S_{30} = 15[2a + 29d] = ?$

$S_{11} = \frac{11}{2}[2a + 10d] = \frac{19}{2}[2a + 18d]$

$22a + 110d = 38a + 342d$

$= 2a + 29d = 0$.

$S_{30} = 15[2a + 29d] \Rightarrow (2a + 29d = 0)$

$15 \times 0 = \underline{0}$

Q. $t_2 + t_9 = t_5 + \boxed{t_6}$

$t_1 + t_{20} = t_6 + \boxed{t_{21}}$

*** Q. If 5th term of an AP is 10 and 10th term of an AP is 0, find 30th term?

$t_5 = 10$, $t_{10} = 0$, $t_{30} = ?$

$a + 4d = 10$ $\qquad$ $a + 9d = 0$.

$a = 10 - 4d$ $\qquad$ $10 = 4d + 9d = 0 \Rightarrow d = -10/5$

$d = 10 - 4(-2)$ $\qquad$ $10 + 5d = 0$ $\qquad$ $\underline{d = -2}$.

$a = 10 + 8 = \underline{18}$

$t_{30} = a + 29d = 18 + 29(-2) = \underline{-40}$

* Geometric Progression : Common ratio

- $t_n = a \times r^{n-1}$
- $S_n = a\frac{(r^n - 1)}{r-1}$ if $r > 1$

$a\frac{(1 - r^n)}{1 - r}$ if $r < 1$.

** Q. 9, 99, 999, 9999, ... t_{100}

Sol^n 10^1-1, 10^2-1, 10^3-1 ... $10^{100}-1$

$\underbrace{10^1+10^2+10^3+\cdots 10^{100}}_{GP:\ a=10,\ r=10} - 100$

$S_n = \frac{a(r^n-1)}{r-1} \Rightarrow \boxed{\frac{10(10^{100}-1)}{9} - 100}$

$\Rightarrow \frac{10^{101}-10-900}{9} = \frac{10^{101}-910}{9}$

** Q. 5, 55, 555, 5555, ... t_{100}

Sol^n $5(1+11+111+\ldots)$ (div. by 9 & mul. by 9)

$\frac{5}{9}(9+99+999+\ldots)$

$\Rightarrow \frac{5}{9}\left[\frac{10^{101}-910}{9}\right]$

Q. 7th term of a GP is 8 times of the 4th term. What will be the first term if its 5th term is 48.

Sol^n: $t_5 = ar^4 = 48$

$t_7 = 8 \times t_4 \Rightarrow ar^6 = 8 \times ar^3$ $\quad r=2$

t_5 $a \times 16 = 48 \Rightarrow a = 3$

$$\boxed{S_\infty = \frac{a}{1-r},\ r<1}$$

Q. $25^{(1/3+1/9+1/27+\cdots\infty)} = ?$

$a = 1/3,\ r = 1/3$

Sol^n $S_\infty = \frac{a}{1-r} = \frac{1/3}{1-1/3} = 1/2$

$25^{(1/2)} = 5$

Q. A ball is thrown from the height of 5000 m on the ground. The ball bounces 4/5 times of its every last bounce. Then calculate the total distance covered by the ball.

Sol^n 9000, 7200 ... ∞

$r = 4/5,\ a = 9000$

$S_\infty = \frac{a}{1-r} \Rightarrow \frac{9000}{1-4/5} = 45000$

Q. Side of Sq. = 16 cm, infinite no. of Sqs. are made by joining the mid points of each side of the square. Calculate the total area of all the infinite no. of Sq.

$a = 256,\ r = 1/2$ $(256, 128, 64, 32, 16 \ldots \infty)$

$(16)^2 = 256,\ (8\sqrt{2})^2 = 128$

as $a_2 = 1/2\, a_1 \therefore a_3 = 1/2\, a_2$

$S_\infty = \frac{256}{1-1/2} = 512\ cm^2$

12/8/25

Q. $11+1003+1005+\cdots$ (100 term); sum = ?

$10^1+1+10^2+3+10^3+5+\cdots 10$

$10^1+10^2+10^3+\cdots(100)$ $\quad \underbrace{1+3+5+\cdots t_{100}}_{n^2 = 10000}$

$a = 10 \quad r = 10 > 1$

$S = 10(10^{100}-1) + 10000$

*** Q. If we consider parts in AP & GP:

AP	GP
3p: $a-d, a, a+d$	3p: $a/r, a, ar$
4p: $a-3d, a-d, a+d, a+3d$	4p: $a/r^3, a/r, ar, ar^3$
5p: $a-2d, a-d, a, a+d, a+2d$	5p: $a/r^2, a/r, a, ar, ar^2$

Q. A no. 20 is divided into four parts that are in AP such that the product of 1st & 4th is to the product of 2nd & 3rd is 2:3. find the largest part.

Sol^n $a-3d + a-d + a+d + a+3d = 20$

$4a = 20$ $\quad \underline{a = 5}$

$5-3d,\ 5-d,\ 5+d,\ 5+3d$ — ①

$\frac{(5-3d)(5+3d)}{(5-d)(5+d)} = \frac{2}{3} \Rightarrow \frac{25-9d^2}{25-d^2} = \frac{2}{3}$

$\underline{d^2 = 1} \Rightarrow d = \pm 1$

Putting $d = \pm 1$ in eq ①,

$5-3d = 2 \quad 5-d = 4 \quad 5+d = 6 \quad 5+3d = 8$

$\therefore$ largest part $5+3d = \underline{8}$

Q. $0.5 + 0.55 + 0.555 + 0.5555 + \ldots$ n term

Sol^n $5(0.1 + 0.11 + 0.111 + \ldots n)$

$\Rightarrow 5/9\,(0.9 + 0.99 + 0.999 + \ldots n)$

$\Rightarrow 5/9\,(1-0.1 + 1-0.01 + 1-0.001 + \ldots n)$

$\Rightarrow 5/9\,[n - (0.1 + 0.01 + 0.001 + \ldots)]$

$\Rightarrow 5/9\,[n - (1/10 + 1/100 + 1/1000 + \ldots)]$

$a = 1/10, \quad r = 1/10 < 1$

$S_n = \frac{a(1-r^n)}{1-r} \Rightarrow \frac{5}{9}\left[n - \frac{\frac{1}{10}\left(1-\frac{1}{10^n}\right)}{1-1/10}\right]$

Q. which term is $1/128$ of the sequence $4, 2, 1 \ldots$

Sol^n $t_n = 1/128, \quad a = 4, \quad r = 1/2$

$t_n = a \times r^{n-1} \Rightarrow 1/128 = 4(1/2)^{n-1}$

$\Rightarrow (1/2)^9 = (1/2)^{n-1} \Rightarrow n-1 = 9 \Rightarrow \underline{n = 10}$

* Q. The Product of 3 consecutive terms of a GP is 216 and the sum of their products in pairs is 156. find the terms

Sol^n a, ar, ar^2

$a \times ar \times ar^2 = 216 \Rightarrow a^3 r^3 = 216 \Rightarrow (ar)^3 = (6)^3 = a = \frac{6}{r}$

$(a \times ar) + (ar \times ar^2) + (a \times ar^2) = 156$

$a^2 r + a^2 r^3 + a^2 r^2 = 156$

$\left(\frac{6}{r}\right)^2 r + \left(\frac{6}{r}\right)^2 r^3 + \left(\frac{6}{r}\right)^2 r^2 = 156$

$a^2 r (1 + r^2 + r) = 156$

$\frac{36}{r^2} \times r(1 + r^2 + r) = 156 \Rightarrow r = 3, 1/3$

	a	ar	ar^2
$r = 3$	$6/3$	$6/3 \times 3$	$6/3 \times 3^2$
$r = 1/3$	6×3	$6 \times 3 \times \frac{1}{3}$	$6 \times 3 \times \left(\frac{1}{3}\right)^2$

$= 2, 6, 18$

$18, 6, 2$

Q. $\frac{1}{1\times2} + \frac{1}{2\times3} + \frac{1}{3\times4} + \ldots + \frac{1}{99\times100}$

Sol^n $\frac{2-1}{1\times2} + \frac{3-2}{2\times3} + \frac{4-3}{3\times4} + \ldots \frac{100-99}{99\times100}$

$= \frac{2}{1\times2} - \frac{1}{1\times2} + \frac{3}{2\times3} - \frac{2}{2\times3} + \frac{4}{3\times4} - \frac{3}{3\times4} + \ldots \frac{100}{99\times100} - \frac{99}{99\times100}$

$= 1 - \frac{1}{100} = \frac{99}{100}$

Q. $\frac{1}{3\times7} + \frac{1}{7\times11} + \frac{1}{11\times15} + \ldots \frac{1}{119\times123}$

Sol^n $\frac{1}{4}\left[\frac{7-3}{3\times7} + \frac{11-7}{7\times11} + \frac{15-11}{11\times15} + \ldots \frac{123-119}{119\times123}\right]$

$\frac{1}{4}\left[\frac{7}{3\times7} - \frac{3}{3\times7} + \frac{11}{7\times11} - \frac{7}{7\times11} + \frac{15}{11\times15} - \frac{11}{11\times15} + \ldots \frac{123}{119\times123} - \frac{119}{119\times123}\right]$

$\frac{1}{4}\left[\frac{1}{3} - \frac{1}{123}\right] = \frac{30}{369} = \boxed{\frac{10}{123}}$

* formula: $\frac{1}{\text{diff. of denominator}}\left[\frac{1}{1^{st}\text{ term of denominator}} - \frac{1}{\text{Last term of denominator}}\right]$

Q. If $a_1 = \frac{1}{2\times5}$ $a_2 = \frac{1}{5\times8}$ $a_3 = \frac{1}{8\times11}$

find $a_1 + a_2 + a_3 + \dots + a_{100}$

Solⁿ: $\frac{1}{3}\left[\frac{1}{2} - \frac{1}{302}\right] = \frac{25}{151}$

$a_{100} = 2, 5, 8 \dots$

$a+(n-1)d = 2 + 99\times3 = 299$

$a_{100} = \frac{1}{299\times302}$

Q. $\frac{1}{1\times4\times7} + \frac{1}{4\times7\times10} + \frac{1}{7\times10\times13} + \frac{1}{10\times13\times16} + \frac{1}{13\times16\times19} + \dots$

Solⁿ: $\frac{7-1}{1\times4\times7} + \frac{10-4}{4\times7\times10} + \frac{13-7}{7\times10\times13} + \frac{16-10}{10\times13\times16} + \frac{19-13}{13\times16\times19}$

$\Rightarrow \frac{\cancel{7}\,1}{1\times4\times7} - \frac{\cancel{4}\,1}{1\times4\times7} + \frac{\cancel{7}\,1}{4\times7\times10} - \frac{\cancel{4}\,1}{4\times7\times10} + \dots \frac{\cancel{19}\,1}{13\times16\times19} - \frac{\cancel{13}\,1}{13\times16\times19}$

$\Rightarrow \frac{1}{6}\left[\frac{1}{1\times4} - \frac{1}{16\times19}\right] =$

★ formula :- $\frac{1}{\text{diff. of 1st \& Last term of 1st denominator}}\left[\frac{1}{\text{Product of 1st \& last of denominator}} - \frac{1}{\text{Product of 1st \& Last of denominator}}\right]$

13/8/25

★ Pigeon hole Principle

If n pigeons are assigned to m pigeon hole and $m<n$ then there's atleast 1 pigeon hole that contain 2 or more pigeon.

→ Extended Pigeon hole Principle.

It states that if n pigeons are assigned to m pigeon hole and n is greater than m ($n>>m$) then one of the pigeon hole must contain atleast $[(n-1)/m]+1$ pigeon.

Q. Prove that if six integers are selected from set of integer $\{3,4,5,6,7,8,9,10,11,12\}$ then there must be two integer whose sum is 15.

3, 12	4, 11	5, 10	6, 9	7, 8

(i) 5 boxes = 5 pigeon hole

(ii) Selected int = Pigeon

① Label 5 boxes with pairs of numbers that sum to 15

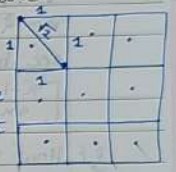
② Every selected integer is placed into the box with the matching label.

According to pigeon hole principle, we now need to place 6 pigeons into 5 pigeon hole. Hence, atleast one box must have two integer.

Q. Prove that if 10 points are placed in 3x3 sq then 2 points must be less than or equal to $\sqrt{2}$ cm apart.

① Divide the sq into 9 sq that are 1x1 cm.

② we have 10 point means 10 pigeons and 9 sq which means 9 pigeon hole. So, by PHP there must be a sq that has more than 1 point or on it within its boundary $\leq \sqrt{2}$ cm apart



Q. Show that atleast two people must have their b'day in the same month if 13 people are assembled in a room.

①

***

Q. Show that if 9 colors are used to paint 100 houses atleast 12 houses are of same colour.

$n = 100$ houses = pigeon, $m = 9$ colors = pigeon hole.

acc to extended pigeon hole principle, $n >> m$

$\left[\frac{n-1}{m}\right] + 1 \quad \frac{100-1}{9} = 11 + 1 = \underline{12}$ //

Q. Seven member of a family have a total of ₹ 2886 in their pocket. show that atleast one of them must have ₹ 414 in their pocket.

$n =$ ₹ 2886 = Pigeon $\quad m$: 7 members = pigeon hole.

acc to EPH, $n >> m$

$\left[\frac{n-1}{m}\right] + 1 \quad \frac{2885}{7} + 1 = 413 + 1 = \underline{414}$ //

Q. How many people must have to guarantee that atleast 9 of them have b'day on the same days of the week

Q. Find the min no of students needed to guarantee that four of them:

i) Born on the same day of week (ii) Born in same.

Q. What is the min no of student req. in a class to be sure that atleast 6 of them will receive the same grade, if there are 5 possible grades.

$m: 7, \; n: ?$

Ans-1) $\left[\frac{n-1}{m}\right] + 1 = 9 \Rightarrow \left[\frac{n-1}{7}\right] = 8 \quad n-1 = 56 \quad n = 56 + 1 = \underline{57}$ //

Ans-3) $\left[\frac{n-1}{m}\right] + 1 = 6 \Rightarrow \frac{n-1}{5} = 5 \quad n-1 = 25 \quad n = 25 + 1 = \underline{26}$ //

Ans-2) $\left[\frac{n-1}{m}\right] + 1 = 4 \Rightarrow \frac{n-1}{7} = 3 \quad n-1 = 21 \quad n = 21 + 1 = \underline{22}$ //

$\left[\frac{n-1}{m}\right] + 1 = 4 \Rightarrow \left[\frac{n-1}{12}\right] = 3 \quad n-1 = 36 \quad n = 36 + 1 = \underline{37}$ //

* GCD: Euclidean Algo

$a = dq + r$

a = dividend, d = divisor, q = quotient, r = remainder.

(*) $\gcd(a, d) = \gcd(d, r)$

Q. $\gcd(34, 12) = 12 \times 2 + 10$

$\gcd(12, 10) = 10 \times 1 + \boxed{2}$

$\gcd(10, 2) = 2 \times 5 + 0$

Q. $\gcd(2322, 654) = 2322 = 654 \times \boxed{3} + \boxed{360}$

$\gcd(654, 360) = 360 \times \boxed{1} + \boxed{294}$

$\gcd(360, 294) = 294 \times \boxed{1} + \boxed{66}$

$\gcd(294, 66) = 66 \times \boxed{4} + \boxed{30}$

$\gcd(66, 30) = 30 \times \boxed{2} + \boxed{6}$

$\gcd(30, 6) = 6 \times \boxed{5} + 0$ $\quad$ Ans = 6 //

Q. $\gcd(1180, 482) = 1180 = 482 \times 2 + 216$

$\gcd(482, 216) = 216 \times 2 + 50$

$\gcd(216, 50) = 50 \times 4 + 16$

$\gcd(50, 16) = 16 \times 3 + 2$

$\gcd(16, 2) = 2 \times 8 + 0$ $\quad$ Ans = 2 //

↳ Extended Euclidean Algo

$\boxed{2 = 1180 \times x + 482 \times y}$ // Bezout's identity.

Q. $\gcd(1180, 482)$

$1180 = 482(2) + 216$

$482 = 216 \times 2 + 50$

$216 = 50 \times 4 + 16$

$50 = 16 \times 3 + 2$

$16 = 2(8) + 0$

reverse $2 = 50 - 16(3)$

$= 50 + 16(-3)$

$\Rightarrow 50 + [216 + 50(-4)](-3)$

$= 50 + 216(-3) + 50(12)$

$= 216(-3) + 50(13)$

$= [482 + 216(-2)](13) + 216(-3)$

$= 216(-29) + 482(13)$

$\Rightarrow 216 - [1180 + 482(-2)](-29) + 482(13)$

$= 1180(-29) + 482(13)$

$\boxed{2 = \underbrace{1180(-29)}_{x} + \underbrace{482(71)}_{y}}$ Bezout's identity

Q. Find the gcd (888, 54)

Solⁿ:- $888 = 54(16) + 24$

$54 = 24(2) + \boxed{6}$ gcd = 6

$24 = 6(4) + 0$

Bezout's identity:- $\boxed{6 = 888(x) + 54(y)}$

$6 = 54 + 24(-2)$

$6 = 54 + [888 + 54(-16)](-2)$

$\boxed{6 = \underbrace{54(33)}_{y} + \underbrace{888(-2)}_{x}}$ Ans

* - Euclidean algo is based on the principle that gcd of two no's doesn't change if the larger no. is replaced by its difference with the smaller number.

- Since this replacement reduces the larger of 2 no's, repeating this process gives smaller pairs of no's until the two no become equal

- By reversing the steps or using the extended euclidean algo. The gcd can be expressed as the linear combination of two no. i.e. the sum of no's multiply by the integers. It is called as Bezout's identity.

* Polynomial & their evolution

$p(x) = g(x) \times q(x) + r(x)$.

Q. Use polynomial long division to perform the division and write the dividend in the form $p(x) = g(x) \times q(x) + r(x)$

$(2x^3 + 7x^2 - 13x - 3) \div (2x - 3)$.

$$
\begin{array}{r|l}
2x-3 & 2x^3 + 7x^2 - 13x - 3 \quad (x^2 + 5x + 1 \\
 & \underline{2x^3 - 3x^2} \\
 & 10x^2 - 13x - 3 \\
 & \underline{10x^2 - 15x} \\
 & 2x - 3 \\
 & \underline{2x - 3} \\
 & \times
\end{array}
$$

$p(x) = (2x - 3) \times (x^2 + 5x + 1) + 0$

Q. $p(x) = 3x^4 - 9x^3 + x^2 + 15 + p$ is completely divisible by $3x^2 - 5$ then find the value of p.

$$
\begin{array}{r|l}
3x^2-5 & 3x^4 - 9x^3 + x^2 + 15 + p \quad (x^2 - 3x + 2 \\
 & \underline{3x^4 - 5x^2} \\
 & -9x^3 + 6x^2 + 15 + p \\
 & \underline{9x^3 - 15} \\
3x^2-5 & 3x^4 - 9x^3 + x^2 + 15x + p \quad (x^2 - 3x + 2 \\
 & \underline{3x^4 - 5x^2} \\
 & -9x^3 + 6x^2 + 15x + p \\
 & \underline{-9x^3 \qquad + 45x} \\
 & 6x^2 + p \\
 & \underline{6x^2 - 10} \\
 & p + 10 = 0
\end{array}
$$

$p = -10$ Ans

* Fundamental principle of counting:-

Q. How many 3 digit no. can be formed from the digit 1, 2, 3, 4, 5. assuming:

(i) repetition is allowed

(ii) repetition not allowed.

Ans →

	H	T	O	
(i)	5	5	5	= 125
(ii)	3	4	5	= 60

Q. How many 3 digit even no. can be formed from 1, 2, 3, 4, 5, 6 assuming (i) repetition allowed (ii) not allowed

Ans →

	H	T	O	
(i)	6	6	3	= 108
(ii)	4	5	3	= 60

Q. How many 5-digit telephone no. can be constructed from digit 0 to 9. if each no. start with 67 and no. repetition is allowed.

Ans → 6 7 8 7 6 $(8 \times 7 \times 6) = 336$

Q. Given 5 flags of diff. colours how many diff. signals can be generated if each signal req. the use of atleast 2 flags one below the other.

Ans → Case 1: 2 flag: $5 \times 4 = 20$

Case 2: 3 flag = $5 \times 4 \times 3 = 60$

Case 3: 4 flag = $5 \times 4 \times 3 \times 2 = 120$

Case 4: 5 flag = $5 \times 4 \times 3 \times 2 \times 1 = 120$

$20 + 60 + 120 + 120 = 320$

Q. Find the value of x: $\frac{1}{6!} + \frac{1}{7!} = \frac{x}{8!}$

Ans → $\frac{1}{6!} + \frac{1}{7 \times 6!} = \frac{x}{8!} \rightarrow \frac{1}{6!}\left[1 + \frac{1}{7}\right] = \frac{x}{8!}$

$\Rightarrow \frac{1}{6!}\left[\frac{8}{7}\right] = \frac{x}{8 \times 7 \times 6!} \rightarrow \frac{8}{7} = \frac{x}{8 \times 7} = x = 64$

* PERMUTATION

- If out of n times, m needs to be arranged then it is given:

$$^nP_m = \frac{n!}{(n-m)!}$$

- If out of n different items, n needs to be arranged. It is $^nP_n = n!$
- If from n objects, m objects is repeating n times. $n!/m!$

Q. How many 4 digit are there with no digit repeated.

→ Th H T O $\quad ^9P_1 \times {}^9P_3$

$9 \times 9 \times 8 \times 7$

Q. How many 4 digit even no. can be formed using a digit 1, 2, 3, 4, 5 with no digit repeated.

→ Th H T O $\quad ^4P_3 \times {}^2P_1$

Q. $^{n-1}P_3 : {}^nP_4 = 1 : 9$

$\rightarrow \frac{^{n-1}P_3}{^nP_4} = \frac{1}{9} \Rightarrow \frac{\frac{(n-1)!}{(n-4)!}}{\frac{n!}{(n-4)!}} = \frac{1}{9} \Rightarrow \frac{(n-1)!}{n \times (n-1)!} = \frac{1}{9} \quad n = 9$

Q. How many words with or without meaning can be formed using all the letters of the word EQUATION, no repetition.

EQUATION = 8 $\rightarrow {}^nP_n = {}^8P_8 = 8!$

Q. word – MONDAY, not repeated.

(i) 4 letters are used at a time: $^6P_4$

(ii) All letters are used $^6P_6 = 6!$

(iii) All used but first is vowel. OA MNDY $= 5! \times 2$

2 ways

Q. Find the no of 8-letter arrangement from the word:

(a) DAUGHTER (b) MISSISSIPPI – 11 words

(i) All vowels occured together (ii) All vowels don't occur together.

(a) DAUGHTER

(i) AVE DGHTR $\Rightarrow 3! \times 6! \Rightarrow 6 \times 6!$

3 ways

(ii) $8! - (6 \times 6!)$

(b) MISSISSIPPI Total possible arrangement $= \frac{11!}{4!\,4!\,2!}$

(i) IIII, MSSSSPP $= \frac{8!}{4! \times 2!} \times 1$

$\frac{4!}{4!} = 1$

(ii) $\frac{11!}{4!\,4!\,2!} - \frac{8!}{4!\,2!}$

Q. PERMUTATIONS:-

(i) Words start with P, end with S. $10!/2!$

(ii) Vowels occur together (AEIOU) PRMTTNS $\Rightarrow \frac{5! \times 8!}{2!}$

5! 7!

(iii) 4 letter b/w P & S. P____S_____

$7 \times 2 \times \frac{10!}{2!}$ — remaining

P&S swap — repetition

* COMBINATION (selection)

- If from n different objects r object needs to be chosen

$\frac{n!}{(n-r)! \times r!}$

- ${}^nC_a = {}^nC_b$ mean $n = a+b$.

Q. If ${}^nC_8 = {}^nC_2$ find ${}^nC_2$

$n = 8+2 = 10 \Rightarrow {}^{10}C_2 = \frac{10!}{8! \times 2!} \Rightarrow \frac{10 \times 9 \times \cancel{8!}}{\cancel{8!} \times 2!} = \frac{90}{2} = 45$

Q. ${}^{2n}C_3 : {}^nC_3 = 12 : 1$

$\frac{{}^{2n}C_3}{{}^nC_3} = \frac{12}{1} \Rightarrow \frac{\overset{2n}{\overline{(2n-3)!}} \times 3!}{\underset{n}{\overline{(n-3)!}} \times 3!} \Rightarrow \frac{(2n)(2n-1)(2n-2)}{(n)(n-1)(n-2)}$

Q. How many chords can be drawn through 21 points on circle.

$\Rightarrow {}^{21}C_2$

2 point

Q. How many ways can a team of 3 boys & 3 girls can be selected from 5 boys & 4 girls.

$\Rightarrow {}^5C_3 \times {}^4C_3$

Q. Determine the no. of 5 card combination out of a deck of 52 cards if there is exactly 1 Ace in each combination.

$\Rightarrow 52 \Rightarrow {}^4C_1 \times {}^{48}C_4$

4 48

Q. A community of 7 has to be form from 9 boys & 4 girls in how many ways this be done if

(i) exactly 3 girls $\Rightarrow {}^4C_3 \times {}^9C_4$

(ii) atleast 3 girls $\Rightarrow {}^4C_3 \times {}^9C_4 + {}^4C_4 \times {}^9C_3$

(iii) atmost 3 girls $\Rightarrow {}^4C_1 \times {}^9C_6 + {}^4C_2 \times {}^9C_5 + {}^4C_3 \times {}^9C_4 + {}^4C_0 \times {}^9C_7$

*Q. If different permutation of all the letter of the word EXAMINATION are listed as in a dictionary. How many words are there in this list before the 1st word starting with E.

$\Rightarrow$ A_____ 10 letters $\Rightarrow \frac{10!}{2! \times 2!}$

Q. In English alphabet, having 5 vowels and 21 consonants. How many words with 2 different vowels and 2 different consonants can be formed.

$\Rightarrow {}^5C_2 \times {}^{21}C_2 \times 4!$

Q. From a class of 25 students 10 are selected for a party. There are 3 students who decided that either all of them will join or none will join. In how many way they can be chosen for the party: ${}^3C_3 \times {}^{22}C_7 + {}^3C_0 \times {}^{22}C_{10}$

Q. In how many ways 5 girl & 3 boys can be seated in a row so that no two boys are together.

→ $5! \times 3! \times {}^6C_3$

Q. Determine the no. of Δ that are formed by selecting points from a set of 15 points, out of which 8 are collinear.

→ ${}^{15}C_3 - {}^{8}C_3$

Q. In how many way 5 software project be alloted to 6 final year student if all the 5 project are not alloted to same student.

→ $6^5 - 6$

Q. What is the no. of ways of chosing 4 cards from a pack of 52 cards. In how many of these:

(i) 4 cards are of same suit.
(ii) 4 cards belongs to different suit.
(iii) All face cards.
(iv) 2 are Red & 2 are black.
(v) Cards are of same colour.

Ans → (i) ${}^{13}C_4 + {}^{13}C_4 + {}^{13}C_4 + {}^{13}C_4 = 4 \times {}^{13}C_4$

(ii) ${}^{13}C_1 \times {}^{13}C_1 \times {}^{13}C_1 \times {}^{13}C_1 = 13^4$

(iii) ${}^{12}C_4$

(iv) ${}^{26}C_2 \times {}^{26}C_2$

(v) ${}^{26}C_4 + {}^{26}C_4 = 2 \times {}^{26}C_4$

Q. In DM paper there are 8 ques in total. In how many way an examiner select if 5 ques in all if 1st is compulsory.

→ ${}^{7}C_4$.

* Division Rule :-

n: Student m: graph $r = \frac{n}{m}$ (size of group).

total no. of ways: $\frac{n!}{(r!)^m \cdot m!}$ (m times)

Q. How many ways to div. 12 students into 3 group of 4 student.

→ $\frac{12}{3! \times 4! \times 4! \times 4!}$

Q. In how many ways we can arrange 2 red tile, 1 yellow tile.

→ $\frac{3!}{2} = 3$

Q. we've 6 clr bricks in total, 4 of them are red, 2 are white. $\frac{6!}{2!\,4!}$

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* Recurrance formula and Recurrance Relation.

A formula which defines any term of a sequence by its previous term is called recursive formula and the relation is called as recurrance relation.

for e.g:- 3, 8, 13, 18, 23

$a_1 = 3$

$a_2 = 8$

a_3

a_4

Q① $S_n = \{2, 2^2, 2^3 \ldots 2^n\}$

② $S = \{3, 7, 11, 15, 19 \ldots\}$

③ $S = \{1, 1, 2, 3, 5, 8, 13 \ldots\}$

① $S_n = \{2, 2^2, 2^3 \cdots 2^n\}$

$S_1 = 2$

$S_2 = 2^2 = 4 = S1 \times 2^1$ $\quad S_n = S_{n-1} \times 2$; $S_1 = 2$, $n \geq 2$

$S_3 = 2^3 = 8 = S2 \times 2$

② $S = \{3, 7, 11, 15, 19 \cdots\}$

$S_1 = 3$ $\quad S_n = S_{n-1} + 4$, S_1

$S_2 = 7 \rightarrow S1 + 4$

⑤ $S = \{1, 1, 2, 3, 5, 8 \cdots\}$

$S_n = S_{n-1} + S_{n-2}$ $\quad S_1 = S_2 = 1$ $\quad n \geq 3$.

→ The recurrance relation can be written in different form

a) $a_{n+3} + 3a_{n+2} + 4a_{n+1} + 8a_n = 0$

b) $y_{n+3} + 3y_{n+2} + 4y_{n+1} + 8y_n = 0$

c) $\mu_{x+3} + 3\mu_{x+2} + 4\mu_{x+1} + 8\mu_n = 0$

d) $f(x+3h) + 3f(x+2h) + 4f(x+h) + 8f(x) = 0$

- Order of recurrance relation.

↳ The difference b/w the highest and lowest subscript of a_n, y_n, μ_x, f_x is called the order of recurrance relation.

- Degree of recurrance relation.

↳ Degree of relation is defined by the highest power of a_n, y_n, μ_x, f_x.

- Linear and Non linear Recurrance Relation.

↳ It's linear if it is of degree one otherwise it is non linear.

- Homogeneous and Non homogeneous

A recurrance relation of the form :

$a_n + a_{n-1} + a_{n-2} + a_{n-3} + \cdots a_{n-k} = f(x)$

if $f(x) = 0$; homogeneous

$f(x) \neq 0$; non homogeneous.

- General Solution :-

a_n = homogeneous solⁿ + particular solⁿ

$= a_h + a_p \rightarrow a^{(h)} + a^{(p)}$

* Solution of linear homogeneous recurrance relation with constant coefficient :-

$C_0 a_n + C_1 a_{n-1} + C_2 a_{n-2} + \cdots C_k a_{n-k} = 0$ —— ①

① Let $a_n = A\alpha^n$, $a_{n-1} = A\alpha^{n-1}$, $a_{n-3} = A\alpha^{n-3}$

② Solve the characterstic eq. and find the root of $\alpha = \alpha_1, \alpha_2, \alpha_3 \cdots \alpha_k$.

③ (i) four distinct real characteristic roots,

Let $\alpha = \alpha_1, \alpha_2$ $\quad \alpha = \alpha_1, \alpha_1, \alpha_2$

$\boxed{a_n = A_1\alpha_1^n + A_2\alpha_2^n}$ $\quad \boxed{a_n = (A_1 + A_2 n)\alpha_1^n + A_3\alpha_2^n}$

(ii) four equal real roots, Let $\alpha = \alpha_1, \alpha_1$

$\boxed{a_n = (A_1 + A_2 n)\alpha_1^n}$

Let $\alpha = \alpha_1, \alpha_1, \alpha_1 \rightarrow \boxed{a_n = (A_1 + A_2 n + A_3 n^2)\alpha_1^n}$

(iii) four complex roots, $\alpha + i\beta$

$\boxed{a_n = \rho^n (A_1 \cos n\theta + A_2 \sin n\theta)}$ where, $\rho = \sqrt{\alpha^2 + \beta^2}$

$\theta = \tan^{-1} \beta/\alpha$

Q. $a_{n+1} - 1.5 a_n = 0$, $n > 0$

Let $a_n = A\alpha^n$, $a_{n+1} = A\alpha^{n+1}$

$A\alpha^{n+1} - 1.5 A\alpha^2 = 0$

$A\alpha^n(\alpha - 1.5) = 0 \rightarrow A\alpha^n \neq 0 \therefore \alpha - 1.5 = 0$ $\quad \alpha = 1.5$

$a_n = A_1 \times \alpha^{n+1} \rightarrow A(1.5)^n$

Q. $a_n = 5a_{n-1} - 6a_{n-2}$, $n \geq 2$, $a_0 = a_1 = 3$

$a_n = A\alpha^n$, $a_{n-1} = A\alpha^{n-1}$, $a_{n-2} = A\alpha^{n-2}$

$a_n - 5a_{n-1} - 6a_{n-2} = 0$

$A\alpha^n - 5A\alpha^{n-1} - 6A\alpha^{n-2}$

$A\alpha^n\left[1 - \frac{5}{\alpha} - \frac{6}{\alpha^2}\right] = 0$

$\alpha^2 - 5\alpha + 6 = 0 \Rightarrow \alpha = 2, 3$

$a_n = A_1(2)^n + A_2(3)^n$

$a_0 = A_1 + A_2 = 3$, $a_1 = 2A_1 + 3A_2 = 3$

$A_1 = 6$, $A_2 = -3$, $\Rightarrow a_n = 6(2)^n - 3(3)^n$ //Ans

Q. $a_n - 6a_{n-1} + 11a_{n-2} - 6a_{n-3} = 0$; $n \geq 3$, $a_0 = 2$, $a_1 = 5$, $a_2 = 15$

$a_n = A\alpha^n$

$A\alpha^n - 6A\alpha^{n-1} + 11A\alpha^{n-2} - 6A\alpha^{n-3} = 0$

$A\alpha^n\left[1 - \frac{6}{\alpha} + \frac{11}{\alpha^2} + \frac{6}{\alpha^3}\right] = 0$

$\alpha^3 - 6\alpha^2 + 11\alpha - 6 = 0$

let $(\alpha = 1)$ $1 - 6 + 11 - 6 = 0 \Rightarrow 0 = 0$ $(\alpha - 1)$

$(\alpha - 1)(\alpha^2 - 5\alpha + 6)$

$\alpha = 1, 2, 3$

	1	-6	11	-6
1		1	-5	6
	1	-5	6	0

$a_n = A_1(1)^n + A_2(2)^n + A_3(3)^n$

$a_0 = A_1 + A_2 + A_3 = 2$ $\quad A_1 = 1$

$a_1 = A_1 + 2A_2 + 3A_3 = 5$ $\quad A_2 = -1$

$a_2 = A_1 + 4A_2 + 9A_3 = 15$ $\quad A_3 = 2$

Q. $a_n + 6a_{n-1} + 12a_{n-2} + 8a_{n-3} = 0$

$\alpha^3 + 6\alpha^2 + 12\alpha + 8 = 0$

$(\alpha + 2)^3 = 0$ $\quad \alpha = -2, -2, -2$

$\boxed{a_n = A_1(-2)^n + A_2(-2)^n + A_3(-2)^n}$

Q. $a_n - 4a_{n-1} + 13a_{n-2} = 0$

$\alpha^2 - 4\alpha + 13 = 0$

$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{4 \pm \sqrt{16 - 52}}{2} = \frac{4 \pm \sqrt{36}}{2} = \frac{4 \pm 6i}{2}$

$\alpha = 2 \pm 3i$, β

$a_n = \rho^n[A_1\cos n\theta + A_2\sin n\theta]$

$\rho = \sqrt{\alpha^2 + \beta^2} = \sqrt{4 + 9} = \sqrt{13}$

$\theta = \tan^{-1}\frac{3}{2}$

$a_n = \sqrt{13}^n[A_1\cos n\tan^{-1}3/2 + A_2\sin n\tan^{-1}3/2]$.

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Q. $y_n + 2y_{n-1} - 15y_{n-2} = 0$; $n \geq 2$ $A y_0 = 0$ $A y_1 = 1$

$y_n = A\alpha^n$, $y_{n-1} = A\alpha^{n-1}$, $y_{n-2} = A\alpha^{n-2}$

$\Rightarrow \alpha^2 + 2\alpha - 15 = 0 \Rightarrow \alpha = 3, \alpha = -5$

$y_n = A_1(\alpha)^n + A_2(\alpha_2)^n$

$= A_1(3)^n + A_2(-5)^n$

If $n = 0$; $y_0 = A_1 + A_2 = 0$ —① $\times 3$

If $n = 1$; $y_1 - 3A_1 - 5A_2 = 1$ —②

$3A_1 + 3A_2 = 0$

$3A_1 - 5A_2 = 1$

$-8A_2 = 1 \Rightarrow A_2 = 1/8$, $A_2 = -1/8$

$y_n = -1/8(3)^n + 1/8(-5)^n$ //Ans

Q. $a_n = a_{n-1} + 2a_{n-1} + 2a_{n-2}$; $n \geq 2$ $a_0 = 0$, $a_1 = 1$

$a_n = A\alpha^n$, $a_{n-1} = A\alpha^{n-1}$, $a_{n-2} = A\alpha^{n-2}$

$\Rightarrow \alpha^2 - \alpha - 2 = 0 \Rightarrow \alpha = 2, -1$

$a_n = A_1(2)^n + A_2(-1)^n$

$a_0 = A_1 + A_2 = 0$ —① $\times 2$

$a_1 = 2A_1 - A_2 = 1$ —②

$3A_1 = 1$

$A_1 = 1/3$ $\quad A_1 = 1/3$, $A_2 = -1/3$.

* Solving Non-homogeneous recurrance relation with constant coefficient :-

Case I $f(n) = \alpha$ (constant)

$a_n = a^H + a^P$

Solving: a^P put (i) $a_n = a_{n-1} = a_{n-2} \cdots a_{n-k} = A$

(ii) find A

(iii) $a^P = A$.

Q. $a_{n+2} - 5a_{n+1} + 6a_n = 2$; $a_0 = 1$, $a_1 = -1$

Order 2 $\rightarrow$ $\alpha^2 - 5\alpha + 6 = 0$

$\alpha = 2, 3$

$a_n = a^H + a^P$

$\boxed{a^H = A_1(2)^n + A_2(3)^n}$

$A - 5A + 6A = 2$ $\rightarrow$ $A = 1$ $\rightarrow$ $a^P = 1$

$a_n = A_1(2)^n + A_2(3)^n + 1$

If $n=0$; $1 + A_1 + A_2 = 1$

If $n=-1$; $1 + 2A_1 + 3A_2 = -1$

$A_1 = 2$, $A_2 = -2$

Sessional **

Q. $y_n - y_{n-1} - 6y_{n-2} = -30$; $y_0 = 20$, $y_1 = -5$.

$\alpha^2 - \alpha - 6 = 0$,

$\alpha = 3, -2$.

$a_n = a^H + a^P$

$a^H = A_1(3)^n + A_2(-2)^n$

Put $y_n = y_{n-1} = y_{n-2} = A$.

$A - A - 6A = -30$

$A = 5$ $\quad y^P = 5$

$y_n = A_1(3)^n + A_2(-2)^n + 5$

$y_0 = A_1 + A_2 + 5 = 20$

$y_1 = 3A_1 - 2A_2 + 5 = -5$

$A_1 = 4$, $A_2 = 11$

Q. Solve : $a_n - 6a_{n-1} + 8a_{n-2} = 3$.

Ans $\alpha - 6\alpha + 8 = 0$

$\alpha = 2, 4$.

$a_n = a^H + a^P$

$a^H = A_1(2)^n + A_2(4)^n$.

Put $a_n = a_{n-1} = a_{n-2} = A$.

$A - 6A + 8A = 3$ $\rightarrow$ $A = 1$

$a_n - 6a_{n-1} + 8a_{n-2} + 1$ $\rightarrow$ Ans $\rightarrow$ $A_1(2)^n + A_2(4)^n + 1$

Q. Solve : $a_n - 5a_{n-1} + 6a_{n-2} = 1$

$\alpha - 5\alpha + 6 = 0$ $\quad \alpha = 2, 3$

$a^H = A_1(2)^n + A_2(3)^n$

Put $a_n = a_{n-1} = a_{n-2} = A$

$A - 5A + 6A = 1$ $\quad 2A = 1$ $\quad A = 1/2$

$a_n - 5a_{n-1} + 6a_{n-2} + 1/2$ Ans.

$a^P = \cancel{A - 5A + 6A + 1/2}$ $A_1(2)^n + A_2(3)^n + 1/2$

Case II $f(n) = p(n)$ - polynomial of degree

(i) let $a_n = A_0 + A_1 n + A_2 n^2 + \cdots$

$= n$

$= n^2$

(ii) Put values of a_n, a_{n-1} ... in given equation.

(iii) Compare the coefficient of like power of n.

(iv) find $A_0, A_1, A_2 \cdots$ and put the value in step ①.

Q. $y_{n+2} - y_{n+1} - 2y_n = n^2$ — ①

$\alpha^2 - \alpha - 2 = 0$ $\quad \alpha = -1, 2$

$y_n = y_H + y_P$ $\rightarrow$ $y_H = A_1(-1)^n + A_2(2)^n$

for finding particular solⁿ;

Let $y_n = A_0 + A_1 n + A_2 n^2$. (Put in eq ①).

$\rightarrow [A_0 + A_1(n+2) + A_2(n+2)^2] - [A_0 + A_1(n+1) + A_2(n+1)^2]$

$- 2[A_0 + A_1 n + A_2 n^2] = n^2$

$\Rightarrow [A_0 + A_1 n + 2A_1 + A_2 n^2 + 4A_2 + A_2 4n] - [A_0 + A_1 n + A_1 + A_2 n^2 + A_2 + 2A_2 n] - 2[A_0 + A_1 n + A_2 n^2] = n^2$

$\Rightarrow$

$\{A_1 - 2A_0 + 3A_2\} + \{2A_2 - 2A_1\}n + \{-2A_2\}n^2 = n^2$

$A_1 - 2A_0 + 3A_2 = 0$

$2A_2 - 2A_1 = 0$

$-2A_2 = 1 \quad \Rightarrow \quad A_2 = -1/2,\ A_1 = -1/2,\ A_0 = -1$

Let $y_n = A_0 + A_1 n + A_2 n^2$

$-1 + (-1/2)n + (-1/2)^2 n$

$y_n = \underbrace{A_1(-1)^n}_{a^n} + A_2(2)^n \underbrace{-1 - n/2 - n^2/2}_{a^p}$ //

Q. $a_{n+2} + 2a_{n+1} - 15a_n = 6n + 10$; $a_0 = 1$, $a_1 = -1/2$

Let $a_n^p = A_0 + A_1 n$

$[A_0 + A_1(n+2)] + 2[A_0 + A_1(n+1)] - 15[A_0 + A_1 n] = 6n + 10$

$[A_0 + 2A_1 + 2A_0 + 2A_1 - 15A_0] + n(A_1 + 2A_1 - 15A_1) = 6n + 10$

Case III: $f(n) = \beta^n \cdot p(n)$

where, $p(n)$ is polynomial funⁿ

β is not a characteristic root.

$a_n^{(p)} = \beta^n \cdot (A_0 + A_1 n + A_2 n^2 \ldots)$

Q. $a_n - 7a_{n-1} + 12a_{n-2} = 2^n$

$a_n = a_n^{(H)} + a_n^{(P)}$

$\boxed{\beta^n = 2^n}$

Let $a_n = A\alpha^n$, $a_{n-1} = A\alpha^{n-1}$, $a_{n-2} = A\alpha^{n-2}$

$\Rightarrow \alpha^2 - 7\alpha + 12 = 0$

$\alpha = 3, 4$

$\boxed{a_n^{(H)} = A_1(3)^n + A_2(4)^n}$ //

Let, $a_n^{(P)} = \beta^n \cdot (A_0 \ldots)$

$= 2^n \cdot (A_0)$ —— (1)

$\Rightarrow 2^n \cdot A_0 - 7(2^{n-1})A_0 + 12(2^{n-2})A_0 = 2^n$

$\Rightarrow 2^n A_0 - \frac{7 \cdot 2^n}{2} A_0 + 3 \cdot 2^n A_0 = 2^n$

$\Rightarrow 2^n[A_0 - 7/2 A_0 + 3A_0] = 2^n$

$\Rightarrow 2A_0 - 7A_0 + 6A_0 = 2$

$A_0 - 7/2 A_0 + 3A_0 = 1 \quad \Rightarrow \quad \boxed{A_0 = 2}$

Putting A_0 in eq 1

$a_n^{(P)} = 2 \cdot 2^n \Rightarrow \boxed{2^{n+1}}$ //

$a_n = A_1(3)^n + A_2(4)^n + 2^{n+1}$ //

Q. $a_n - 7a_{n-1} + 12a_{n-2} = n \times 2^n$ find $a_n^{(P)}$

$a_n^{(P)} = \beta^n \cdot (A_0 + A_1 n \ldots)$

$= 2^n(A_0 + A_1 n)$

$\Rightarrow 2^n(A_0 + A_1 n) - 7[2^{n-1}(A_0 + A_1(n-1))] + 12[2^{n-2}(A_0 + A_1(n-2))] = n \times 2^n$

$\Rightarrow 2^n(A_0 + A_1 n) - 7/2[A_0 2^n + A_1 n \cdot 2^n - A_1 \cdot 2^n]$

$+ 3[A_0 \cdot 2^n + n \cdot 2^n A_1 - 2 \cdot 2^n A_1] = n \times 2^n$

$\Rightarrow n \cdot 2^n\left(A_1 - \frac{7}{2}A_1 + 3A_1\right) + 2^n\left(A_0 - 7/2 A_0 + 7/2 A_1 + 3A_0 - 6A_1\right)$

$= n \cdot 2^n$

$A_1 - 7/2 A_1 + 3A_1 = 1 \rightarrow A_1 = 2$

$A_0 - 7/2 A_0 + 7/2 A_1 + 3A_0 - 6A_1 = 0 \rightarrow A_0 = 10$

$a_n^{(P)} = 2^n (A_0 + A_1 n) \rightarrow 2^n (10 + 2n)$ //

Case IV

$f(x) = \beta^n \cdot p(x)$

Where $p(x)$ is the polynomial of degree S.

β^n is a characteristic root

$\boxed{a_n^{(P)} = n^t \beta^n (A_0 + A_1 n)}$

Where t is the multiplicity of root

$\alpha = 2, 2, 3$

Q. $a_n - 7a_{n-1} + 12 a_{n-2} = 4^n$

$\alpha^2 - 7\alpha + 12 = 0 \rightarrow \alpha = 3, 4$

$a_n^{(H)} = A_1(4)^n + A_2(3)^n$

Let $a_n^{(P)} = n 4^n (A_0)$

$\rightarrow n 4^n A_0 - 7[(n-1)4^{n-1} A_0] + 12[(n-2)4^{n-2}(A_0)] = 4^n$

$\rightarrow 4^n [n A_0 - \frac{7}{4}(n-1)A_0 + \frac{3}{4}(n-2)A_0] = 4^n$

$\rightarrow A_0 [n - 7/4(n-1) + 3/4(n-2)] \rightarrow (\cancel{n} - \cancel{n}/4 + 1/4 + 3\cancel{n}/4 - 3/2) = 1$

$\rightarrow 4A_0 = 16 \rightarrow A_0 = 4$

$a_n^{(P)} = n 4^n (4)$ //

*** Q. $a_n - 4a_{n-1} + 4a_{n-2} = (n+1) 2^n$

$\alpha^2 - 4\alpha + 4 = 0 \rightarrow \alpha = 2, 2$

$a_n^{(H)} = (A_1 + A_2 n) 2^n$

Let $a_n = n^2 2^n (A_0 + A_1 n)$

$\rightarrow n^2 2^n (A_0 + A_1 n) - 4[(n-1)^2 2^{n-1} (A_0 + A_1(n-1))]$

$+ 4[(n-2)^2 2^{n-2} (A_0 + A_1(n-2))] = n \cdot 2^n + 2^n$

$\rightarrow 2^n (A_1 n + A_0) n^2 - 4 \cdot 2^{n-1} [A_1(n-1) + A_0](n-1)^2 +$

$4 \cdot 2^{n-2} [A_1(n-2) + A_0](n-2)^2 = n \cdot 2^n + 2^n$

$\rightarrow 2^n (A_1 n + A_0) n^2 - 2(n^2 + 1 - 2n) \cdot 2^n (A_1 n - A_1 + A_0) +$

$(n^2 + 4 - 4n) \cdot 2^n (A_1 n - 2A_1 + A_0) = n \cdot 2^n + 2^n$

$\rightarrow 2^n [A_1 n^3 + A_0 n^2 + (-2n^2 - 2 + 4n)(A_1 n - A_1 + A_0)$

$+ (n^2 + 4 - 4n)(A_1 n - 2A_1 + A_0)] = n \cdot 2^n + 2^n$

$\rightarrow 2^n [A_1 n^3 + A_0 n^2 - 2A_1 n^3 + 2A_1 n^2 - 2A_0 n^2 - 2A_1 n + 2A_1 - 2A_0$

$+ 4A_1 n^2 - 4A_1 n + 4A_0 n + A_1 n^3 - 2A_1 n^2 + A_0 n^2 + 4A_1 n$

$- 8A_1 + 4A_0 - 4A_1 n^2 + 8A_1 n - 4A_0 n] = n \cdot 2^n + 2^n$

$\rightarrow 2^n [6A_1 n + 6A_1 + 2A_0] = n \cdot 2^n + 2^n$

$\rightarrow A_1 = 1/6, A_0 = 1$ //

** Compulsary Question (part A)

Q. Q. If $a|bc$ and $\gcd(a, b) = 1$, then prove that $a|c$.

Q. State pigeon hole principle.

Q. If 5 term of ap is 10 and 10th term is 0. find 30th term.

Q. What is the generating function of $\{1, 1, 1, 1, \ldots\}$

Q. Consider the binary relation $R = \{(x, y), (x, z), (z, x), (z, y)\}$ on the set $\{x, y, z\}$. which

(a) R is Symmetric and not anti symetric

(b) R is not Symmetric but anti symmetric

(c) R is both Symmetric and anti Symmetric.

(d) R is neither Symmetric

Section - A

Q2) (a) $A \times (B \cup C) = (A \times B) \cup (A \times C)$

(b) $(A \cap B) - C = (A - C) \cap (B - C)$

(c) $(A \cup B)' = A' \cap B'$

(B) Explain multiset and various ~~explanation~~ operations on multiset with the help of example

Q3. (A) In A Survey of 60 people it was found that 25 ppl. read news week magzine, 26 read time, 26 read fortune

9 read both newsweek & fortune, 11 read both news & week, time, 8 read both time & fortune, 3 read all the magzine.
- find the no. of people who atleast read one of these
- who read exactly one magzine
- Draw the venn diagram.

(B) Let S be the set of all points in a plane. let R be a relation such that for any two points a & b $(a,b) \in R$ if b is within 2 cm from A. Check whether R is an equivalence relation or not.

Q4. (A) Let $A = \{1,2,3\}$ Consider the power set P(A) of A list all the element of P(A). Let R be the relation of set inclusion of P(A) that is for $(x,y) \in R$ if $x \subseteq y$ prove that: P(A) is a POSET w.r.t. R.

(B) Explain the properties of Relation with example.

Section - B

Q5. Solve the following: $a_n - 4a_{n-1} + 4a_{n-2} = (n+1) * 2^n$

Q6. $y_n - y_{n-1} - 6y_{n-2} = -30$ given that $y_0 = 20$, $y_1 = -5$.

Q7. $a_{n+2} - 2a_{n+1} + a_n = 2^n$ using the method of generating function $a_0 = 2$, $a_1 = 1$.

Q8 (A) How many permutation can be made out of BASIC.
a) How many of these begin with B.
b) ends with C
c) b & C occupy the end places.

(B) Show that if 9 colours are used to paint 100 houses atleast 2 houses will be of the same colour.

Generating function:

Sequence: $a_0, a_1, a_2 \ldots a_k$

$G(x) = a_0 x^0 + a_1 x^1 + a_2 x^2 \ldots a_k x^k = \sum_{k=0}^{\infty} a_k x^k$

*** Q. 1, 1, 1 . . . , or $a_k = 1$

$\rightarrow G(x) = 1 * x^0 + 1 * x^1 + 1 * x^2 \ldots \infty$

$G(x) = 1 + x + x^2 \ldots \infty$

$a = 1$, $r = x$

$S_\infty = \frac{a}{1-r} = \frac{1}{1-x} \rightarrow \boxed{G(x) = \frac{1}{1-x}}$

Q. 1, 2, 3, 4, 5 . . . i.e. $a_k = k+1$

$G(x) = 1 \times x^0 + 2 \times x^1 + 3 \times x^2 + 4 \times x^3 + \ldots$

$= 1 + 2x + 3x^2 + 4x^3 + \ldots$ —— ①

$x \cdot G(x) = x + 2x^2 + 3x^3 + 4x^4 + \ldots$ ②

$(1-x)G(x) = 1 + x + x^2 + x^3$

$(1-x)(G(x)) = \frac{1}{1-x} \rightarrow \boxed{G(x) = \frac{1}{(1-x)^2}}$

Q. 1, -1, 1, -1 . . . $a_k = (-1)^k$

$G(x) = 1 \times x^0 + (-1) \times x^1 + (1) \times x^2 + (-1) \times x^3 \ldots$

$= 1 - x + x^2 - x^3 + \ldots$

$a = 1$, $r = -x$ $\boxed{S_\infty = 1/1+x}$

Q. 1, 1, 0, 1, 1, 1 . . .

$G(x) = 1 \times x^0 + 1 \times x^1 + 0 \times x^2 + 1 \times x^3 \ldots$

$= 1 + x + x^3 + x^4 + \ldots$

$= (1 + x + x^2 + x^3 + \ldots) - x^2$

$\boxed{\frac{1}{1-x} - x^2}$

Q. $0, 1, 2, 3 \ldots a_k = k$

$G(x) = 0 \times x + 1 \times x^1 + 2 \times x^2 + 3 \times x^3$

$= 0 + x + 2x^2 + 3x^3 + \ldots$

$= x[1 + 2x + 3x^2 + \ldots]$

$= \dfrac{x}{(1-x)^2}$

Q. $G(x) = \dfrac{3}{1-x} \Rightarrow a_x = 3$ $\quad$ Q. $G(x) = \dfrac{x}{(1-x)^2} + \dfrac{3}{1-x} = a(x)$

$\Rightarrow$ formula:

	a_k	$G(x)$
1.	1	$1/1-x$
2	$(-1)^k$	$1/1+x$
3.	c	$c/1-x$
4.	$k+1$	$1/(1-x)^2$
5.	k	$x/(1-x)^2$
6.	a^k	$1/1-ax$
7.	$(-a)^k$	$1/1+ax$
8.	k^2	$x(1+x)/(1-x)^3$
9.	$k \cdot a^k$	$ax/(1-ax)^2$

*** Solution of Recurrance Relation by Generating function.

1. In given eq. multiply by x^n if variable is n.
2. Take Submission from $n = 1$ to ∞ if one initial condition is given and $n = 2$ to ∞ if two initial conditions are given.
3. write each submation in terms of $G(x)$ or closed form
4. Put value of each Σ and find $G(x)$.
5. find partial fraction of $G(x)$
6. write a_n

Q. Solve: $a_n - 2a_{n-1} - 3a_{n-2} = 0$; $a_0 = 3, a_1 = 1$ using generating function.

$\sum_{n=2}^{\infty} a_n x^n - 2\sum_{n=2}^{\infty} a_{n-1} x^n - 3\sum_{n=2}^{\infty} a_{n-2} x^n = 0$ — (1)

$\sum_{n=2}^{\infty} a_n x^2 = a_2 x^2 + a_3 x^3 + a_4 x^4 + \ldots \infty$

$= G(x) - a_0 - a_1 x \Rightarrow \boxed{G(x) - 3 - x}$ — (2)

$a_0 = a_n : 2 \to \infty$
$a_1 = a_n : 1 \to \infty$
$a_{n+2} = 0 \to \infty$
$a_{n+1} = 1 \to \infty$

$\sum_{n=2}^{\infty} a_{n-1} x^n = a_1 x^2 + a_2 x^3 + a_3 x^4 + \ldots$

$= x[a_1 x + a_2 x^2 + a_3 x^3 \ldots]$

$= x[a(x) - a_0] = x[G(x) - 3]$ — (3)

$\sum_{n=2}^{\infty} a_{n-2} x^n = a_0 x^2 + a_1 x^3 + a_2 x^4 + \ldots$

$= x^2 [a_0 + a_1 x + a_2 x^2 + \ldots]$

$x^2 G(x)$ — (4)

Putting eq. 2, 3, 4 in eq 1.

$\Rightarrow G(x) - 3 - x - 2(x(G(x) - 3)) - 3(x^2 G(x)) = 0$

$G(x)[1 - 2x - 3x^2] - 3 - x + 6x = 0$

$G(x)[1 - 2x - 3x^2] = -5x + 3$

$G(x) = \dfrac{-5x+3}{1-2x-3x^2} \Rightarrow \dfrac{-3+5x}{3x^2+2x-1} \Rightarrow \dfrac{5x-3}{(x+1)(3x-1)}$

$\Rightarrow \dfrac{5x-3}{(x+1)(3x-1)} = \dfrac{A}{(x+1)} + \dfrac{B}{(3x-1)}$

let $x+1 = 0 \Rightarrow x = -1 \Rightarrow -8/-4 = 2$ — (A)

$3x - 1 = 0 \Rightarrow x = 1/3 \Rightarrow -1$ — (B)

$\Rightarrow \dfrac{2}{(x+1)} + \dfrac{(-1)}{(3x-1)} \Rightarrow 2(-1)^n + \dfrac{1}{1-3x}$

$\Rightarrow \boxed{a_n = 2(-1)^n + (3)^n}$

Q. $a_n - a_{n-1} - 2a_{n-2} = 0$; $a_0 = 0$; $a_1 = 1$ using generating fun^n

$\Rightarrow \sum_{n=2}^{\infty} a_n x^n - \sum_{n=2}^{\infty} a_{n-1} x^n - 2\sum_{n=2}^{\infty} a_{n-2} x^n = 0$ — (1)

$\Rightarrow \sum_{n=2}^{\infty} a_n x^2 = a_2 x^2 + a_3 x^3 + a_4 x^4 + \ldots$

$= G(x) - a_0 - a_1 x = \boxed{a(x) - x}$ — (2)

$$\sum_{n=2}^{\infty} a_{n-1} x^n = a_1 x^2 + a_2 x^3 + a_3 x^4 + \cdots$$
$$= x[a_0 + a_1 x + a_2 x^2 \cdots]$$
$$= \boxed{xG(x)} \quad \text{—} (3)$$

$$\sum_{n=2}^{\infty} a_{n-2} x^n = a_0 x^2 + a_1 x^3 + a_2 x^4 \cdots$$
$$= x^2[a_0 + a_1 x + a_2 x^2] = \boxed{x^2 G(x)} \quad \text{—} (4)$$

Put (2)(3)(4) in eq (1)

$G(x) - x - xG(x) - 2x^2 G(x) = 0$

$G(x)[1 - x - 2x^2] = x$

$$G(x) = \frac{x}{1-x-2x^2} = \frac{x}{-(2x+1)(x+1)} = \frac{A}{1-2x} + \frac{B}{1+x}$$

$A = 1/3,\ B = -1/3$

$$G(x) = \frac{1}{3(1-2x)} - \frac{1}{3(1+x)}$$

$=$

f. $a_n = 3a_{n-1} + 2;\ a_0 = 1$

$$\sum_{n=1}^{\infty} a_n x^n = 3\sum_{n=1}^{\infty} a_{n-1} x^n + 2\sum_{n=1}^{\infty} x^n \quad \text{—} (1)$$

$$\rightarrow \sum_{n=1}^{\infty} a_n x^n = a_1 x + a_2 x^2 + \cdots \rightarrow \boxed{G(x) - 1} \quad \text{—} (2)$$

$$\rightarrow \sum_{n=1}^{\infty} a_{n-1} x^n = a_0 x + a_1 x^2 + a_2 x^3 + \cdots$$
$$= x[a_1 + a_1 x + a_2 x^2 \cdots] = \boxed{x(G(x))} \quad \text{—} (3)$$

$$\rightarrow \sum_{n=1}^{\infty} x^n = x + x^2 + x^3 + \cdots = \boxed{\left[\frac{1}{1-x}\right]} \quad \text{—} (4)$$

Put eq 2,3,4 in (1).

$G(x) - 1 = 3[xG(x)] + 2\left[\frac{x}{1-x}\right]$

$\rightarrow G(x)[1 - 3x] = 1 + 2x/1-x$

$$\rightarrow G(x) = \frac{1+2x}{(1-x)(1-3x)} \rightarrow \frac{A}{1-x} + \frac{B}{1-3x}$$

$\rightarrow A = -1,\ B = 2 \rightarrow \frac{-1}{1-x} + \frac{2}{1-3x}$

$\rightarrow \boxed{a_n = -1 + 2 \cdot 3^n}$

g. $a_{n+2} - 2a_{n+1} + a_n = 2^n;\ a_0 = 2,\ a_1 = 1$

$$\sum_{n=0}^{\infty} a_{n+2} \cdot x^n - 2\sum_{n=0}^{\infty} a_{n+1} + \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} 2^n \cdot x^n \quad \text{—} (1)$$

$$\sum_{n=0}^{\infty} a_{n+2} x^n = a_2 x^0 + a_3 x^1 + a_4 x^2 + \cdots$$
$$= 1/x^2[a_2 x^2 + a_3 x^3 + a_4 x^4 + \cdots]$$
$$= 1/x^2[G(x) - a_0 - a_1 x] = 1/x^2[G(x) - 2 - x] \quad \text{—} (2)$$

$$\sum_{n=0}^{\infty} a_{n+1} \cdot x^n = a_1 x^0 + a_2 x^1 + a_3 x^2 + \cdots$$
$$= 1/x[a_1 x + a_2 x^2 + a_3 x^3 + \cdots] = 1/x[G(x) - 2] \quad \text{—} (3)$$

$$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots = G(x) \quad \text{—} (4)$$

$$\sum_{n=0}^{\infty} 2^n \cdot x^n = 1 + 2x + 4x^2 + 8x^2 \rightarrow \underline{1/1-2x}$$

$\rightarrow 1/x^2[G(x) - 2 - x] - 2/x[G(x) - 2] + G(x) = 1/1-2x$

$\rightarrow G(x) - 2 - x - 2x(G(x) - 2) + G(x) \cdot x^2 = x^2/1-2x$

$\rightarrow G(x)[1 - 2x + x^2] = 2 - 3x + x^2/1-2x$

$\rightarrow G(x)[(1-x)^2] = 2 - 3x + x^2/1-2x$

$$\rightarrow G(x) = \frac{2}{(1-x)^2} - \frac{3x}{(1-x)^2} + \frac{x^2}{(1-x)^2(1-2x)}$$

$$\rightarrow \frac{x^2}{(1-2x)(1-x)^2} = \frac{A}{(1-2x)} + \frac{B}{1-x} + \frac{C}{(1-x)^2}$$

let $1 - 2x = 0$

$1 = 2x \rightarrow x = 1/2 \rightarrow A = \underline{1}$

let $(1-x)^2 = 0$

$x = 1 \qquad C = -1$

$$\frac{1}{(1-2x)} + \frac{B}{1-x} - \frac{1}{(1-x)^2} \rightarrow \frac{(1-x)^2 + B(1-2x)(1-x) - 1(1-2x)}{(1-2x)(1-x)^2}$$

$x^2 = x^2(1 + 2B) - x(3B) + B \rightarrow \boxed{B = 0}$

$$\rightarrow G(x) = \frac{2}{(1-x)^2} - \frac{3x}{(1-x)^2} + \frac{1}{1-2x} - \frac{1}{(1-x)^2}$$

$$= \frac{1}{(1-x)^2} - \frac{3x}{(1-x)^2} + \frac{1}{1-2x} \rightarrow a_n = (n+1) - 3n + (2)^n$$